

Multiplication of Complex Numbers as Rotation



Mathematics | JEE Main & Advanced

Don't Memorize. Visualize.

Multiplying by a complex number w turns the arrow by $\arg w$ and scales its length by $|w|$. The lengths multiply and the angles add, so one multiplication can rotate any point about the origin.

ROTATE

Turn the arrow about the origin. Its length stays the same; only its direction changes. Anticlockwise is the positive direction, as for the argument.

SCALE

Stretch or shrink the arrow along its own line. Its direction stays the same; only its length changes.

Scaling by 2 doubles the length, and scaling by $\frac{1}{2}$ halves it.

As in the previous lesson, a complex number $z = x + iy$ is drawn as an arrow from the origin to the point (x, y) on the Argand plane.

Multiply by i : a quarter turn

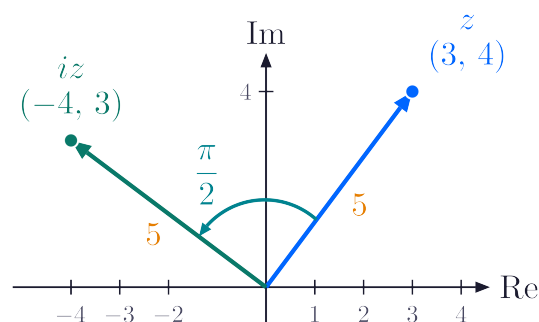
$$i(3 + 4i) = 3i + 4i^2 = 3i - 4 = -4 + 3i.$$

The point $(3, 4)$ moves to $(-4, 3)$. Both arrows have length 5, so nothing was stretched. The arrow has turned by $\frac{\pi}{2}$ anticlockwise, from the first quadrant into the second.

Multiplying by i turns every point

$$i(x + iy) = ix + i^2y = -y + ix$$

The point (x, y) always moves to $(-y, x)$: a quarter turn anticlockwise about the origin. Multiplying by $-i$ sends (x, y) to $(y, -x)$: a quarter turn clockwise.



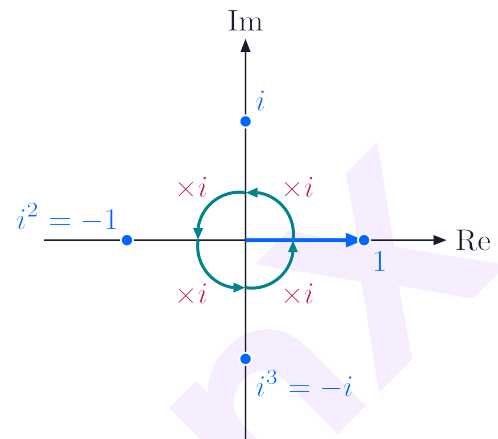
Powers of i : why $i^2 = -1$

Start with the arrow to 1. Each multiplication by i is one more quarter turn. Two quarter turns make a half turn, and a half turn takes 1 to -1 . That is why $i^2 = -1$.

The powers of i : a cycle of four

$$i^1 = i, \quad i^2 = -1, \quad i^3 = -i, \quad i^4 = 1$$

Four quarter turns make a full turn, so the cycle repeats. For every whole number k : $i^{4k} = 1$, $i^{4k+1} = i$, $i^{4k+2} = -1$ and $i^{4k+3} = -i$.



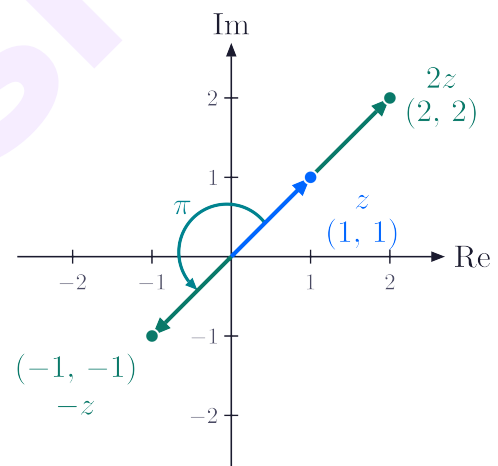
Multiply by a real number

A positive real number only scales

A positive real number k has argument 0, so it does not turn the arrow. The arrow gets longer if $k > 1$ and shorter if $0 < k < 1$. For example, $2(1 + i) = 2 + 2i$ is twice as long as $1 + i$ and points the same way.

A negative real number adds a half turn

A negative real number k has argument π : it turns the arrow by half a turn and scales it by $|k|$. Multiplying by -1 is a half turn alone, because $-1 = i \times i$ is two quarter turns: $-(1 + i) = -1 - i$, the point $(-1, -1)$.



The general rule: lengths multiply, angles add

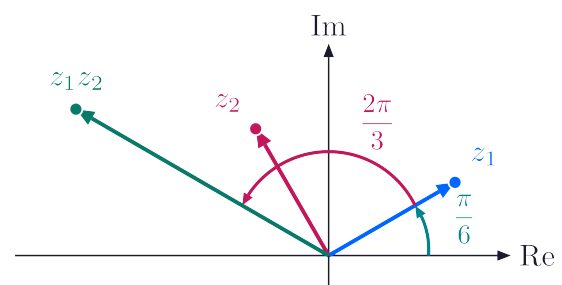
The rule

For any two nonzero complex numbers z_1 and z_2 ,

$$|z_1 z_2| = |z_1| |z_2|, \quad \arg(z_1 z_2) = \arg z_1 + \arg z_2 \quad (\text{up to a multiple of } 2\pi).$$

Take $z_1 = \sqrt{3} + i$, the point $(\sqrt{3}, 1)$, with length 2 and angle $\frac{\pi}{6}$, and $z_2 = -1 + \sqrt{3}i$, the point $(-1, \sqrt{3})$, with length 2 and angle $\frac{2\pi}{3}$.

Turn the arrow of z_1 by $\frac{2\pi}{3}$ and stretch it by 2: the product $z_1 z_2$ has length $2 \times 2 = 4$ and angle $\frac{\pi}{6} + \frac{2\pi}{3} = \frac{5\pi}{6}$.



Check with algebra

$$(\sqrt{3} + i)(-1 + \sqrt{3}i) = -\sqrt{3} + 3i - i + \sqrt{3}i^2 = -2\sqrt{3} + 2i$$

Its length is $\sqrt{12 + 4} = 4$. It is in the second quadrant with $\alpha = \tan^{-1} \frac{2}{-2\sqrt{3}} = \frac{\pi}{6}$, so its argument is $\pi - \frac{\pi}{6} = \frac{5\pi}{6}$. The same point.

Why it works: the polar form

Write $z_1 = r_1(\cos \theta_1 + i \sin \theta_1)$ and $z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$. Open the brackets and use $i^2 = -1$:

$$\begin{aligned} z_1 z_2 &= r_1 r_2 [(\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) + i(\sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2)] \\ &= r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]. \end{aligned}$$

This is the polar form of a number with length $r_1 r_2$ and angle $\theta_1 + \theta_2$.

Euler form and the rotation rule

Euler's formula

The arrow of length 1 at angle θ ends at the point $(\cos \theta, \sin \theta)$. That number is written

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

So every nonzero complex number is $z = r e^{i\theta}$ with $r = |z|$ and $\theta = \arg z$, and $z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}$: the powers of e add, so the angles add.

Rotation rule

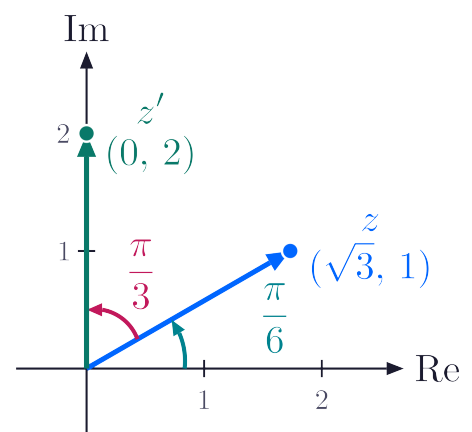
To turn z about the origin through an angle θ :

$$\text{anticlockwise: } z' = z e^{i\theta}, \quad \text{clockwise: } z' = z e^{-i\theta}.$$

$e^{i\theta}$ has length 1, so it turns without scaling. To turn by θ and also multiply the distance from the origin by $k > 0$, multiply by $k e^{i\theta}$.

Turn $z = \sqrt{3} + i$ (length 2, angle $\frac{\pi}{6}$) anticlockwise by $\frac{\pi}{3}$. The new angle is $\frac{\pi}{6} + \frac{\pi}{3} = \frac{\pi}{2}$, so the arrow lands on the positive imaginary axis at $(0, 2)$. With the rule:

$$(\sqrt{3} + i) \left(\frac{1}{2} + \frac{\sqrt{3}}{2} i \right) = \frac{\sqrt{3}}{2} + \frac{3}{2} i + \frac{1}{2} i + \frac{\sqrt{3}}{2} i^2 = 2i.$$

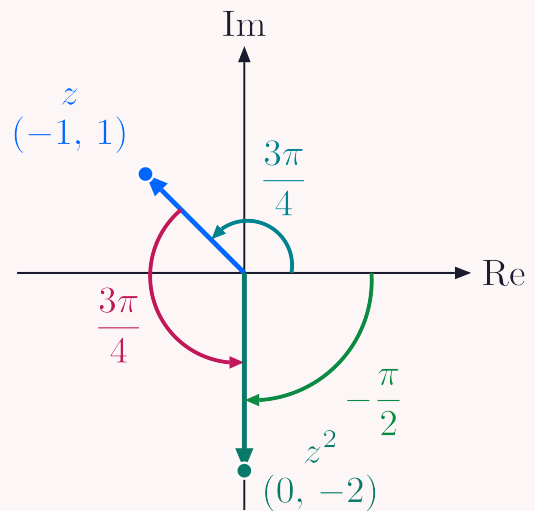


CAREFUL

Adding the angles can take you out of the principal range $(-\pi, \pi]$. Take $z = -1 + i$, with $\arg z = \frac{3\pi}{4}$. Then $z^2 = (-1 + i)^2 = 1 - 2i + i^2 = -2i$.

Adding the angles gives $\frac{3\pi}{4} + \frac{3\pi}{4} = \frac{3\pi}{2}$, and the arrow really does turn that far. But $\frac{3\pi}{2} > \pi$, so it is not the principal argument: $\arg(z^2) = \frac{3\pi}{2} - 2\pi = -\frac{\pi}{2}$.

After adding the angles, always check the range. If the sum is outside $(-\pi, \pi]$, add or subtract 2π ; one correction is always enough.



Solved example

The point $2 + i$ is turned anticlockwise about the origin through $\frac{\pi}{4}$, and its distance from the origin is multiplied by $\sqrt{2}$. Find the new point.

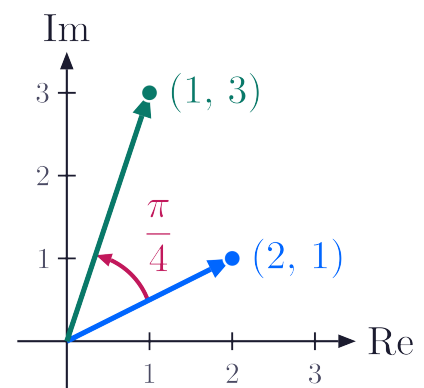
The multiplier. One number w must turn by $\frac{\pi}{4}$ and stretch by $\sqrt{2}$, so $|w| = \sqrt{2}$ and $\arg w = \frac{\pi}{4}$:

$$w = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) = \sqrt{2} \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} i \right) = 1 + i.$$

Multiply. $(2 + i)(1 + i) = 2 + 2i + i + i^2 = 1 + 3i$.

Check. $|1 + 3i| = \sqrt{10} = \sqrt{2} \times \sqrt{5} = \sqrt{2} |2 + i|$.

The new point is $1 + 3i$, that is $(1, 3)$.



Key formulas

Lengths multiply: $|z_1 z_2| = |z_1| |z_2|$

Angles add: $\arg(z_1 z_2) = \arg z_1 + \arg z_2$

Quarter turn: $i(x + iy) = -y + ix$

Powers of i : $i^2 = -1$, $i^4 = 1$

Euler: $e^{i\theta} = \cos \theta + i \sin \theta$

Turn by θ anticlockwise: $z' = z e^{i\theta}$

Turn by θ clockwise: $z' = z e^{-i\theta}$

Turn by θ , scale by $k > 0$: $z' = z k e^{i\theta}$

The angle rule holds up to a multiple of 2π : after adding, bring the answer back into $(-\pi, \pi]$.

Practice set

Easy to hard. Give every argument as the principal argument, in radians. Sketch each point first: the picture tells you where it went. The answers start on the next page.

- Q1.** Find $i(2 - 5i)$. How has the point moved?
- Q2.** Find $-i(3 + 4i)$.
- Q3.** $|z_1| = 3$, $\arg z_1 = \frac{\pi}{6}$, $|z_2| = 2$, $\arg z_2 = \frac{\pi}{3}$. Find $z_1 z_2$.
- Q4.** Find the modulus and the argument of $(1 + i)(1 + \sqrt{3}i)$.
- Q5.** Turn $1 + \sqrt{3}i$ clockwise through $\frac{\pi}{2}$ about the origin.
- Q6.** Turn $2 + 2i$ anticlockwise through $\frac{\pi}{4}$ about the origin.
- Q7.** Let $z = -\sqrt{3} + i$. Find $\arg(z^2)$. Is it equal to $2 \arg z$?
- Q8.** $2 + i$ is one vertex of a square whose centre is the origin. Find the other three vertices.



Watch the animated lesson on YouTube

Multiplication of Complex Numbers as Rotation

Scan the code to open the JEEVisionX channel on YouTube. This lesson is in the Complex Numbers playlist.

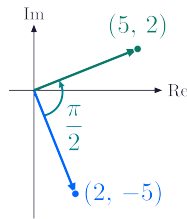
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Answers

Q1

$$5 + 2i$$

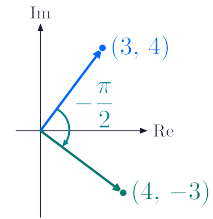
$i(2 - 5i) = 2i - 5i^2 = 5 + 2i$. The point $(2, -5)$ moves to $(5, 2)$: a quarter turn anticlockwise.



Q2

$$4 - 3i$$

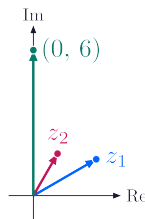
$-i(3 + 4i) = -3i - 4i^2 = 4 - 3i$: a quarter turn clockwise.



Q3

$$6i$$

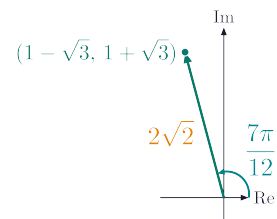
Length $3 \times 2 = 6$, angle $\frac{\pi}{6} + \frac{\pi}{3} = \frac{\pi}{2}$, so $z_1 z_2 = 6(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}) = 6i$.



Q4

modulus $2\sqrt{2}$,
argument $\frac{7\pi}{12}$

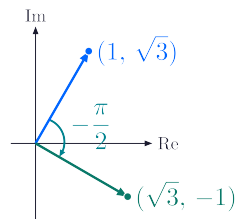
Lengths $\sqrt{2}$ and 2 multiply to $2\sqrt{2}$. Angles $\frac{\pi}{4} + \frac{\pi}{3} = \frac{7\pi}{12}$, already in $(-\pi, \pi]$. Check: the product is $(1 - \sqrt{3}) + (1 + \sqrt{3})i$, in the second quadrant.



Q5

$$\sqrt{3} - i$$

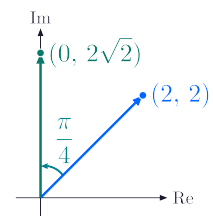
Multiply by $e^{-i\pi/2} = -i$: $-i(1 + \sqrt{3}i) = -i - \sqrt{3}i^2 = \sqrt{3} - i$.



Q6

$$2\sqrt{2}i$$

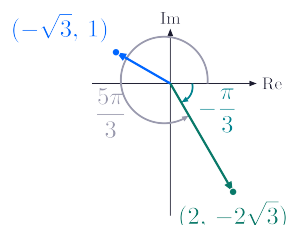
$|2 + 2i| = 2\sqrt{2}$ and its angle is $\frac{\pi}{4}$. Turning by $\frac{\pi}{4}$ gives angle $\frac{\pi}{2}$ with the same length: the point $(0, 2\sqrt{2})$.



Q7

$\arg(z^2) = -\frac{\pi}{3}$, and no.

$\arg z = \frac{5\pi}{6}$, so $2 \arg z = \frac{5\pi}{3}$ (the grey turn), outside $(-\pi, \pi]$. $z^2 = 3 - 2\sqrt{3}i + i^2 = 2 - 2\sqrt{3}i$, in the fourth quadrant, with argument $\frac{5\pi}{3} - 2\pi = -\frac{\pi}{3}$.



Q8

$$-1 + 2i, -2 - i, 1 - 2i$$

The vertices of a square centred at the origin are quarter turns of one another, so multiply $2 + i$ by i , i^2 and i^3 .

